

POLVIFIDEL: EVALUATING AND IMPROVING THE POLITICAL FIDELITY IN VLM IMAGE DESCRIPTIONS

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ABSTRACT

Political images contain distinctive actors, symbols, and relationships that existing caption evaluation metrics fail to capture. Reference-based metrics inherit annotators’ knowledge and biases, while embedding-based measures overlook whether captions reflect politically meaningful content. I introduce POLVIFIDEL, the first reference-free metric for assessing the political fidelity of image captions. POLVIFIDEL uses domain-specific face recognition and political object detection to extract a structured set of political entities from an image and measures how well a caption mentions or meaningfully describes them. I further show that supplying these detected entities as auxiliary prompts enables VLMs to generate more politically faithful captions without finetuning. Across multiple proprietary and open-source VLMs, POLVIFIDEL aligns more closely with human political-grounding judgments than existing metrics and highlights systematic failures in describing low-salience politicians and rare symbols. The framework offers a scalable, domain-grounded approach for evaluating and improving political image captioning.

1 INTRODUCTION

Image descriptions generated by vision–language models (VLMs) are increasingly used in political communication research to extract visual features and uncover latent themes in large-scale image corpora (Steinert-Threlkeld & Xie, 2025; Lokmanoglu & Walter, 2025). Yet we still know little about whether these captions faithfully capture politically salient visual content. For example, a VLM may caption a photo of Donald Trump wearing a MAGA hat as “a man wearing a red hat”—accurate but politically uninformative. Such failures illustrate that extracting political meaning from images requires identifying political actors, symbolic objects, and the relationships among them. I refer to this requirement as political grounding.

Existing evaluation practices offer limited insight into political grounding. Reference-based metrics inherit human annotators’ knowledge gaps, subjective decisions, and ideological priors—especially problematic in domains where political context deeply shapes what annotators choose to mention. Conversely, visual fidelity metrics relying on general-purpose object detectors overlook domain-specific cues essential for interpreting political scenes, such as party-specific apparel, or protest symbols. As a result, current approaches cannot reliably assess whether VLM descriptions truly reflect the political content present in an image. This gap is particularly consequential as political scientists increasingly rely on VLM-generated captions at scales—tens or hundreds of thousands of images—where manual evaluation is infeasible.

Beyond measuring accuracy, it is equally important to determine whether political grounding in captions can be improved, and how. Fine-tuning VLMs is largely infeasible for most political scientists due to data, expertise, and compute requirements. Therefore, we need evaluation tools and improvement strategies that are model-agnostic and do not require finetuning, yet meaningfully enhance the political informativeness of captions.

This paper makes two contributions in response to these challenges. First, I introduce POLVIFIDEL, a domain-adapted visual fidelity metric that extends VIFIDEL (Madhyastha et al., 2019)

using specialized political object and actor detectors, and penalizes hallucinated entities lacking visual support. Second, I evaluate whether auxiliary visual prompting—providing cropped object images, textual detection summaries, or annotated images—improves political grounding in VLM descriptions. This tests whether combining accurate political object detection with VLM relational reasoning yields more faithful political captions without fine-tuning. Together, these contributions advance both the evaluation and improvement of VLM-generated captions for political imagery.

2 RELATED WORK

2.1 CAPTION EVALUATION: REFERENCE-BASED VS. VISUAL FIDELITY METRICS

Image caption evaluation has traditionally relied on reference-based metrics, such as BLEU (Papineni et al., 2002), METEOR (Banerjee & Lavie, 2005), ROUGE (Lin, 2004), CIDEr (Vedantam et al., 2014), and SPICE (Anderson et al., 2016), which measure textual similarity between a generated caption and human-written references. While widely adopted in computer vision, these metrics inherit human annotators’ subjective decisions, knowledge constraints, and political priors—limitations that are especially salient in political imagery where annotators may omit or misidentify politically meaningful cues (Berger et al., 2025). Another line of work focuses on image-based visual fidelity metrics. VIFIDEL (Madhyastha et al., 2019) evaluates consistency between captioned entities and automatically detected objects, while CHAIR (Rohrbach et al., 2018) penalizes hallucinations unsupported by the underlying image. Region-grounding metrics similarly assess whether tokens correspond to specific image regions. However, all existing approaches rely on general-purpose detectors, which often fail to recognize political actors, partisan symbols, or protest-related objects. POLVIFIDEL extends this line of work by incorporating domain-specific actor and symbol detectors into VIFIDEL’s framework and by directly penalizing hallucinated political entities, offering a domain-adapted measure of how faithfully captions reflect political content.

2.2 VLMS VS. TASK-SPECIFIC VISION MODELS FOR ACTORS, OBJECTS, AND RELATIONS

Although VLMS are trained on massive vision–language corpora, their ability to take over classic vision tasks such as detection and recognition remains limited. Across multiple evaluations, VLMS consistently underperform task-specific vision models in basic perception, including object detection, fine-grained recognition, and bounding-box localization—tasks for which dedicated detectors such as DETR (Carion et al., 2020) and Grounding-DINO (Liu et al., 2024) remain state of the art. Empirical studies also document that VLMS frequently hallucinate entities or omit visually present actors and objects when used without structured visual inputs, a pattern well-documented in hallucination analyses (Rohrbach et al., 2019). Yet once entities are provided, VLMS exhibit strong relational and interaction reasoning capabilities. Recent advances in human–object interaction (HOI) and visual relationship detection rely on this division of labor: nearly all state-of-the-art HOI systems use a detection backbone to localize humans and objects, and then employ a VLM-based module to infer interactions or relations (Kang et al., 2024; Zhao et al., 2023; Lei et al., 2025; Yamada et al., 2025). This pattern suggests a complementary strength: detectors provide reliable grounding for actors and objects, while VLMS excel at inferring the relations among them—motivating the hybrid prompting strategy examined in this study.

2.3 CONTROLLING AND IMPROVING VLM CAPTIONS WITHOUT FINETUNING

Early controllable captioning models such as Show, Control and Tell (Cornia et al., 2019) and MAGIC (Su et al., 2022) demonstrated that caption content can be guided through structured inputs—such as control sequences or semantic attributes—even though these models required task-specific training. Modern VLMS extend this idea while keeping most model parameters frozen. Models such as InstructBLIP (Dai et al., 2023), LLaVA-NeXT (Li et al., 2024), and Qwen-VL (Bai et al., 2023) show that caption grounding can be improved through input conditioning—including vision features extracted by frozen encoders, multimodal prefix prompts, or interleaved image–text exemplars—without fine-tuning large pretrained components. These developments demonstrate that caption behavior can be meaningfully influenced at inference time, a property especially valuable in settings where fine-tuning is impractical. Building on this insight, this study tests whether supply-

ing structured political object information can similarly improve the political grounding of VLM-generated captions.

3 PROPOSED METHOD

I present a framework for evaluating and improving the political grounding of image descriptions by leveraging outputs from domain-specific vision models. I first describe how political actor and object representations are constructed using specialized vision detectors (Section 3.1), then introduce POLVIFIDEL, a domain-adapted visual fidelity metric that operationalizes political grounding (Section 3.2), and finally outline an auxiliary-information prompting strategy that uses detected objects to improve VLM-generated descriptions (Section 3.3).

3.1 POLITICAL ENTITY REPRESENTATION

I construct a political entity representation for each image–caption pair, consisting of politically meaningful actors and objects detected in the image and mentioned in the caption.

3.1.1 POLITICAL ACTOR AND OBJECT DETECTION

I define a political entity set $\mathcal{O}$ containing all detected political actors and objects, obtained using domain-specific face recognition and object detection models.

Given an input image I , the political entity set is defined as $\mathcal{O} = \{(o_i, b_i, e_i)\}_{i=1}^M$ where each o_i is a political entity (either actor or object) b_i its bounding box, and e_i its embedding representation. Unlike VIFIDEL, this representation incorporates domain-specific object detectors and also include human actors.

3.1.2 LINGUISTIC ENTITY EXTRACTION AND ALIGNMENT

From a candidate caption $\mathcal{C}$, I extract noun phrases and entity mentions: $\mathcal{W} = \{w_j : j = 1, \dots, N\}$.

To align textual mentions with visual entities, I avoid CLIP-based semantic matching—commonly used in prior visual-fidelity metrics—because such embeddings often conflate politically meaningful distinctions within broad object categories. Instead, I generate a dictionary-based matching system, where each detector class o_i is associated with a large set of paraphrases, synonyms, and descriptive variants obtained using LLMs.

The final alignment matrix $A \in \{0, 1\}^{M \times N}$ is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if } w_j \text{ matches } o_i \text{ via dictionary,} \\ 0 & \text{otherwise.} \end{cases}$$

3.2 POLVIFIDEL: A POLITICAL VISUAL-FIDELITY METRIC

POLVIFIDEL evaluates whether a caption accurately identifies the political actors and objects visible in an image. It extends VIFIDEL in two key ways: (1) it replaces the generic object detector with our domain-specific political actor and object detectors (Section 3.1.1), and (2) it introduces an explicit hallucination penalty, reflecting the cost of fabricating political actors or symbols.

POLVIFIDEL consists of three components.

3.2.1 OBJECT RECALL

$$R_{\text{obj}} = \frac{|\{i : \exists j, A_{ij} = 1\}|}{M}$$

which measures the share of detected political entities mentioned in the caption.

3.2.2 HALLUCINATION PENALTY

$$\text{HAL} = \frac{\sum_{w_j \in W_{\text{hall}}} \alpha_j}{\sum_{w_j \in W} \alpha_j},$$

where importance weights α_j assign higher penalties to salient political actors and symbols.

3.2.3 FINAL METRIC

$$\text{POLVIFIDEL} = R_{\text{obj}} \cdot (1 - \text{HAL})^\beta, \quad \beta \geq 1$$

where $\beta \geq 1$ controls the severity of hallucination penalties. When $\beta = 1$, hallucinations reduce fidelity linearly; for larger β , hallucinations are penalized more aggressively.

This formulation preserves the grounding logic of VIFIDEL while adding domain-specific precision and hallucination control.

3.3 IMPROVING VLM DESCRIPTIONS VIA AUXILIARY OBJECT INFORMATION

I investigate whether VLM-generated image descriptions can be improved by providing structured political entity information alongside the original image. I evaluate three prompting strategies:

1. *Visual Aux*: cropped object images
2. *Textual Aux*: detector outputs provided as structured text (e.g., “3 police officers, 2 American flags”)
3. *Annotated Aux*: detector outputs overlaid on the original image with textual tags

These strategies supply explicit grounding cues while allowing the VLM to perform relational and situational reasoning—an area where VLMs often outperform task-specific HOI or relation-detection models. By combining high-precision vision-based entity detection with VLM relational inference, I expect obtain descriptions with significantly better political grounding than either component alone.

4 EXPERIMENT PLAN

Dataset: I evaluate our method on 1,200 political news images curated from The New York Times. To capture a diverse political visual domain, we stratify images across (1) elite vs. mass events, (2) Republican vs. Democratic actors and contexts, and (3) local vs. national political levels. Each image is paired with an editorial caption, which I treat as a noisy but informative reference text for computing standard reference-based metrics.

Models and Prompting Conditions: I benchmark five VLMs—two proprietary (GPT-4o, Gemini 2.5) and three open-source (LLaVA-1.6, Intern-S1, Qwen3-VL)—under four prompting conditions: *Textual Aux*: standard caption generation without additional inputs, *Visual Aux*, *Textual Aux*, and *Annotated Aux* as mentioned in Section 3.3. Each model produces captions for all images under all prompting conditions, yielding 24,000 total captions.

Procedure: For each caption, I compute two visual metrics—VIFIDEL, and our proposed POLVIFIDEL—and five standard reference-based metrics—CIDEr, BLEU, ROUGE, METEOR, and SPICE—using the original NYT caption as reference. To obtain an external assessment of grounding quality, I recruit human annotators who rate captions on a five-point scale, focusing exclusively on visual fidelity: whether the caption accurately describes visible political actors, objects, and relations. They are explicitly asked to exclude fluency and stylistic preferences. Each caption receives three independent ratings, averaged to produce a final human fidelity score. I compute Spearman correlations between human fidelity scores and each automatic metric across all models and prompting conditions. This analysis assesses whether POLVIFIDEL better tracks human judgments than other metrics, and whether prompting VLMs with auxiliary political detections improves the political grounding of generated captions.

5 EXPECTED RESULTS

Evaluation: I expect POLVIFIDEL to exhibit substantially higher correlation with human judgments than existing metrics. Because POLVIFIDEL directly measures whether captions correctly identify political actors and objects while penalizing hallucinations, it should closely mirror how human evaluators assess whether a description faithfully reflects visible political content.

Improvements in political grounding via auxiliary inputs: I also expect that providing auxiliary object information to VLMs will consistently improve caption fidelity, reflected in both higher human ratings and higher POLVIFIDEL scores. While no single prompting strategy—cropped object images, textual object lists, or annotated images—is expected to dominate, all should outperform the no-auxiliary baseline. I further anticipate heterogeneous gains across images: captions for low-salience events or lesser-known political figures should benefit most, as VLMs exhibit more frequent errors in these settings.

Improvements for downstream visual analysis: Finally, I expect that increases in caption fidelity will propagate to downstream analyses. More faithful captions should enable higher-quality visual feature extraction, leading to more coherent and politically meaningful topic-modeling outputs based on VLM-generated descriptions. Thus, improvements in political grounding are likely to yield measurable gains in downstream representational accuracy, demonstrating the broader utility of enhancing caption fidelity.

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